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Centre Number

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Student Number

2018

Year 12 Extension 2 Mathematics task 1

Thursday 13th December 2018

General Instructions

- Reading time – 5 minutes
- Working time – 40 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations

Total Marks:
25

Section I – 17 marks

- Allow about 12 minutes for this section

Section II – 13 marks

- Allow about 13 minutes for this section

Section 1 Complex Numbers	Question 1	/7
	Question 2	/5
Section 2 Polynomial	Question 1	/5
	Question 2	/5
	Question 3	/3
Total		/25

Section 1 (12 marks)

Write each question in a separate booklet.

Marks

Question 1

- (i) Write $\frac{1+i\sqrt{3}}{1+i}$ in the form $x + iy$, where x and y are real. 2
- (ii) Write $\frac{1+i\sqrt{3}}{1+i}$ in modulus-argument form. 2
- (iii) Hence find the exact value of $\cos \frac{\pi}{12}$. 1
- (iv) Hence calculate $\left(\frac{1+i\sqrt{3}}{1+i}\right)^{12}$. 2

Question 2

- a) On an Argand diagram shade the region containing all points representing complex numbers, z , such that
- $$1 \leq |z + 1| \leq 2 \quad \text{and} \quad \operatorname{Re}(z + 1) + \operatorname{Im}(z) > 0$$
- 3
- b) Find the range of possible values of $\arg z$? 2

Section 2 (13 marks)

Write each question in a separate booklet.

Marks

Question 1

- (i) Show that $(-1 + i\sqrt{3})$ is a zero of $3x^3 + 5x^2 + 10x - 4 = 0$. 3
- (ii) Hence, find the other roots. 2

Question 2

- (i) The polynomial $P(x) \equiv 3x^4 - 5x^3 - x^2 + 6x + 5$ is divided by $(x^2 + x - 2)$. Find the remainder. 3
- (ii) If $A(x) \equiv 3x^4 - 5x^3 - x^2 + ax + b$ is divisible by $(x^2 + x - 2)$, find the values of a and b . 2

Question 3

On an Argand diagram, the points A, B and C represent the complex numbers $(1+2i)$, $(5+2i)$ and $(3+7i)$ respectively. D is the mid-point of AB.

- (i) Use vectors to prove that AB is perpendicular to CD. 2
- (ii) Hence, find the length of CD. 1

END OF ASSESSMENT

Validation test-solution

Question 1

$$(i) \quad \frac{(1+i\sqrt{3})(1-i)}{(1+i)(1-i)} = \frac{1-i+i\sqrt{3}+\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2} + \frac{(\sqrt{3}-1)i}{2} \quad (1)$$

$$(ii) \quad 1 + \sqrt{3}i = 2cis\frac{\pi}{3}, \quad 1 + i = \sqrt{2}cis\frac{\pi}{4}$$

$$\rightarrow \frac{1+i\sqrt{3}}{1+i} = \sqrt{2}cis\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$$

$$= \sqrt{2}cis\frac{\pi}{12} = \frac{1+\sqrt{3}}{2} + \frac{(\sqrt{3}-1)i}{2} \quad \text{from (1)}$$

$$(iii) \quad \therefore \sqrt{2} \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2} \quad \rightarrow \quad \cos \frac{\pi}{12} = \frac{1+\sqrt{3}}{2\sqrt{2}}$$

$$(iv) \quad \left(\frac{1+i\sqrt{3}}{1+i}\right)^{12} = \left(\sqrt{2}cis\left(\frac{\pi}{3} - \frac{\pi}{4}\right)\right)^{12}$$

$$= 2^6 cis\pi = -64$$

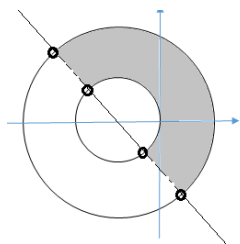
2 marks

2 marks

1 mark

Question 2

a)



$Re(z+1) + Im(z) > 0$ is the region $x+1+y > 0$

3 marks:

1 for each correct boundary and region.
Must have open circle.

b) Intercepts between $(x+1)^2 + y^2 = 1$ and $y = -(x+1)$

$$(-y)^2 + y^2 = 1 \quad \rightarrow \quad y = \pm \frac{1}{\sqrt{2}} \quad \text{and} \quad x = \frac{-1-\sqrt{2}}{\sqrt{2}}, \frac{1-\sqrt{2}}{\sqrt{2}}$$

$$\therefore \min. arg = \tan^{-1} \frac{\frac{1}{\sqrt{2}}}{\frac{-1-\sqrt{2}}{\sqrt{2}}} = -22.5^\circ = -\frac{7\pi}{8} \quad \text{max. arg} = \tan^{-1} \frac{\frac{1}{\sqrt{2}}}{\frac{1-\sqrt{2}}{\sqrt{2}}} = 67.5^\circ = -\frac{5\pi}{8}$$

$$\therefore -\frac{5\pi}{8} < arg z < \frac{7\pi}{8} \quad \begin{matrix} \text{(2nd region)} \\ \text{(3rd region)} \end{matrix}$$

2 marks working and correct answers

Question 3

$$\begin{aligned} \text{(i)} \quad w &= 1 - \frac{2i}{z} \\ &= 1 - \frac{2i\bar{z}}{|z|^2} \\ &= \left(1 - \frac{2y}{x^2 + y^2}\right) - \frac{2ix}{x^2 + y^2} \end{aligned}$$

(ii) w is undefined when $z = 0$

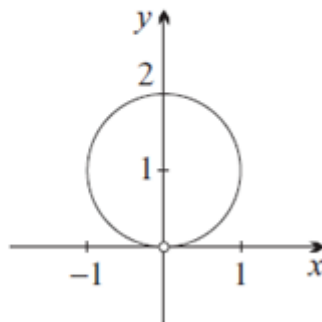
(iii) Since w is pure imaginary,

$$\operatorname{Re}(w) = 0$$

$$\text{so } x^2 + y^2 - 2y = 0$$

$$\text{or } x^2 + (y - 1)^2 = 1$$

Thus the locus is the unit circle with centre $z = i$, omitting the origin.



2 marks

1 mark

2 marks